Personal Learning Journey – Karunarathna H.R.A.

Student	Student ID	Group	Member	Role
Karunarathna H.R.A.	IT24102885	2026-AI-33	Member 4	Evaluation & Recommendation

My assigned role is to translate the selected models into an honest account of grade error, risk-flag reliability and possible stakeholder value. The main lesson from this stage is that a good-looking average score can conceal important missed support cases.

Contribution and evidence

Notebook 04 loads the two already-fitted pipelines, verifies their data and model metadata, and predicts the same 79 held-out students without refitting. It reports MAE, RMSE and R^2, residual plots, errors by observed grade group, risk metrics, confusion matrices and descriptive sex/school checks. Complete results remain in the notebook, while individual case views show up to ten rows to keep the presentation readable.

A finding that shaped the recommendation

The main model has test MAE 3.095, RMSE 3.871 and R^2 0.269. Its rule finds only 10 of the 27 students below grade 10, misses 17 and produces 11 false positives. This makes recall particularly important: an unflagged student cannot be assumed to need no help. The G1/G2 model has MAE 1.170 and finds all 27 in this sample, but produces 10 false positives and requires later-period information.

Learning and critical judgement

A simple G2-unchanged prediction has lower MAE, 1.000, than the later-stage forest. G2 < 10 misses one student with six false positives, so the forest catches one extra student at the cost of four extra flags. This comparison shows why recommendations should discuss workload and missed assistance rather than a single score. Residuals also reveal overprediction of low grades and large errors among zero-grade students. Perfect observed recall in a small test sample is not a future guarantee.

Team handoff and next steps

The recommendations connect Member 3's frozen models to staff-reviewed support, such as checking current performance and discussing tutoring or study planning. The model should not determine grading, discipline or denial of assistance. A new cohort is needed for further validation; the current test set is already used. Input availability, zero-grade causes, the small two-school sample and uncertain subgroup estimates remain limitations. My next priorities would be defining a prediction date and evaluating the practical support workload with stakeholders.

Personal reflection

The result I checked most carefully was the later-stage comparison. I recounted the confusion counts for the forest and for the G2 < 10 rule and confirmed that the forest's one extra correct flag costs four extra reviews. That changed my recommendation from preferring the forest to piloting it against the simple G2 rule. The limitation I would explain first to teaching staff is that the main model misses 17 of 27 at-risk students, so an unflagged student must still be offered help.

Evidence and AI use

Evidence: notebooks/04_evaluation_recommendations.ipynb; src/evaluation_utils.py. I used an AI assistant to explain evaluation metrics, including precision, recall, specificity and confusion matrices. I recounted the confusion counts myself from the Notebook 04 predictions before writing the recommendation. AI also helped draft this report; its figures were checked against the notebook outputs.